

Determination of astrophysical ${}^7\text{Be}(p, \gamma){}^8\text{B}$ reaction rate from the ${}^7\text{Li}(\text{d}, \text{p}){}^8\text{Li}$ reaction

DU XianChao¹, GUO Bing^{1*}, LI ZhiHong¹, PANG DanYang^{2,3}, LI ErTao⁴ & LIU WeiPing^{1†}

¹China Institute of Atomic Energy, P. O. Box 275(10), Beijing 102413, China;

²School of Physics and Nuclear Energy Engineering, Beihang University, Beijing 100191, China;

³International Research Center for Nuclei and Particles in the Cosmos, Beihang University, Beijing 100191, China;

⁴College of Physics Science and Technology, Shenzhen University, Shenzhen 518060, China

Received Month date, Year; accepted Month date, Year

The ${}^7\text{Be}(p, \gamma){}^8\text{B}$ reaction not only plays a central role in the evaluation of solar neutrino fluxes but also could affect the evolution of the first stars. The neutron asymptotic normalization coefficient (ANC) for the virtual decay of ${}^8\text{Li}_{g.s.} \rightarrow {}^7\text{Li} + \text{n}$ is derived through the distorted wave Born approximation (DWBA) analysis of the ${}^7\text{Li}(\text{d}, \text{p}){}^8\text{Li}$ angular distribution. According to charge symmetry, the square of proton ANC for ${}^8\text{B}_{g.s.} \rightarrow {}^7\text{Be} + \text{p}$ is then extracted. We have also deduced the astrophysical S-factors and reaction rates for direct capture in ${}^7\text{Be}(p, \gamma){}^8\text{B}$ at energies of astrophysical relevance. The astrophysical S-factor at zero energy for direct capture, $S_{17}(0)$, is derived to be 20.5 ± 3.3 eV b in good agreement with the most recent recommended value. We have also calculated the contribution of the 1^+ and 3^+ resonances for the S-factor and reaction rate. This work provides an independent examination to the preexisting results of the ${}^7\text{Be}(p, \gamma){}^8\text{B}$ reaction.

${}^7\text{Li}(\text{d}, \text{p}){}^8\text{Li}$, $E_d = 12$ MeV, DWBA analysis, mirror system, ${}^7\text{Be}(p, \gamma){}^8\text{B}$, astrophysical S-factors and reaction rates

PACS number(s): 25.60.Je; 21.10.Jx; 25.40.Lw; 26.20.+f

Citation: Du X C, Guo B, Li Z H, et al. Determination of astrophysical ${}^7\text{Be}(p, \gamma){}^8\text{B}$ reaction rate from the ${}^7\text{Li}(\text{d}, \text{p}){}^8\text{Li}$ reaction. Sci China-Phys Mech Astron, 2014, 57: 1–6, doi:

1 Introduction

The ${}^7\text{Be}(p, \gamma){}^8\text{B}$ reaction plays a crucial role in constraining the properties of neutrino oscillations and solar composition through solar-neutrino experiments [1, 2] since ${}^8\text{B}$ is the major source of the high-energy solar neutrinos. Recent experiments were able to determine the neutrino flux emitted from ${}^8\text{B}$ decay with a precision of 9% [3], while the theoretical predictions have uncertainties of the order of 20% [4]. Theoretical neutrino flux critically depends on the ${}^7\text{Be}(p, \gamma){}^8\text{B}$ reaction.

误差应比这个小吧，比如看 Alderberger 的综述文章

The ${}^7\text{Be}(p, \gamma){}^8\text{B}$ reaction also plays an important role in the evolution of the first stars. The first generation of stars

formed at the end of the cosmic dark ages, which marked the key transition from a homogeneous and simple universe to a highly structured and complex one [5]. A classic question on the evolution of these supermassive stars is whether they contributed any significant material to later generations of stars by supernova explosions. In 1986, the evolution of non-rotating supermassive stars was studied with a hydrodynamic code KEPLER [6] where concluded that these stars will collapse into black holes without experiencing a supernova explosion. This is because the triple alpha process ($3\alpha \rightarrow {}^{12}\text{C}$) does not produce sufficient amounts of CNO seed nuclei so that the hot CNO cycle and rp -process are unable to generate the nuclear energy enough to explode the stars. In 1989, the

*Corresponding author (email: guobing@ciae.ac.cn)

†Contributed by LIU WeiPing

rap-processes was suggested as alternative paths which would permit these stars to bypass the 3α process and to yield the CNO material [7]. The ${}^7\text{Be}(p, \gamma){}^8\text{B}$ reaction is one of the key reactions in these rap-processes.

Due to its importance, the ${}^7\text{Be}(p, \gamma){}^8\text{B}$ reaction has increasingly attracted both theoretical and experimental studies in past years. As for the experiment, the direct measurements were performed with the proton beam on the ${}^7\text{Be}$ targets [8–11], or with the radioactive ${}^7\text{Be}$ beam on the H_2 target [12, 13]. In addition, indirect measurement were carried out through Coulomb dissociation method [14–18] and the transfer reaction method [19–21]. As for the theoretical side, several methods were utilized, including the R-matrix parametrization [22], the potential model [23, 24], the microscopic cluster models [25–27], and the ab initio calculations [28, 29].

In this work, we derived the neutron asymptotic normalization coefficient (ANC) for the virtual decay ${}^8\text{Li} \rightarrow {}^7\text{Li} + n$ by reanalysing the exiting angular distribution at $E_d=12$ MeV (the experimental data are taken from Ref. [30]) based on the distorted wave Born approximation (DWBA) analysis, and we then deduced the proton ANC for the virtual decay ${}^8\text{B} \rightarrow {}^7\text{Be} + p$ based on charge symmetry of mirror nuclei. The ${}^7\text{Be}(p, \gamma){}^8\text{B}$ reaction astrophysical S-factors and reaction rates of direct capture into ground state of ${}^8\text{B}$ are then obtained. We have also calculated the contribution from the 1^+ and 3^+ resonances.

应该说一下电荷对称性在多大程度满足，并引用参考文献！

2 The neutron ANCs for virtual decay ${}^8\text{Li} \rightarrow {}^7\text{Li} + n$

For a peripheral transfer reaction, the neutron ANC of ${}^8\text{Li} \rightarrow {}^7\text{Li} + n$ can be extracted by comparison of the experimental differential cross sections with the DWBA calculations,

$$\left(\frac{d\sigma}{d\Omega}\right)_{\text{exp}} = \sum_{j_i j_f} (C_{l_i j_i}^d)^2 (C_{l_f j_f}^{8\text{Li}})^2 R_{l_i j_i l_f j_f}, \quad (1)$$

where

$$R_{l_i j_i l_f j_f} = \frac{\sigma_{l_i j_i l_f j_f}^{\text{DWBA}}}{(b_{l_i j_i}^d)^2 (b_{l_i j_i}^{8\text{Li}})^2}, \quad (2)$$

$\left(\frac{d\sigma}{d\Omega}\right)_{\text{exp}}$ and $\sigma_{l_i j_i l_f j_f}^{\text{DWBA}}$ are the experimental and theoretical differential cross sections, respectively. $C_{l_i j_i}^d$, $C_{l_f j_f}^{8\text{Li}}$, $b_{l_i j_i}^d$ and $b_{l_i j_i}^{8\text{Li}}$ represent nuclear and corresponding single-particle ANCs for virtual decays $d \rightarrow p + n$ and ${}^8\text{Li} \rightarrow {}^7\text{Li} + n$, respectively. l_i , j_i and l_f , j_f mean the orbital and total angular momenta of the transferred neutron in initial and final nuclei d and ${}^8\text{Li}$, respectively. $R_{l_i j_i l_f j_f}$ is model independent under the condition of peripheral transfer reaction.

Table 1 Optical potential parameters used in the calculations, where V and W are in MeV, r and a in fm.

Set No.	1		2	
channel	Entrance	Exit	Entrance	Exit
V	118	45	78	45
r_0	0.886	1.32	0.9	1.32
a	0.907	0.57	0.95	0.57
W_s	5.8	11	30	11
r_i	1.57	1.32	0.9	1.32
a_i	0.777	0.345	0.8	0.345
V_{so}	5.8	5.0	5.8	5.0
r_{so}	0.886	1.32	0.9	1.32
a_{so}	0.907	0.57	0.95	1.57

In the present calculation, we used the finite-range DWBA code fresco [31] to analyze the angular distributions. C_d^2 is taken to be 0.76 fm^{-1} from Ref. [32]. There are two components, corresponding to the neutron transfers to the $1p_{1/2}$ and $1p_{3/2}$ orbits, which contribute to the ${}^7\text{Li}(d, p){}^8\text{Li}_{g.s.}$ reaction. The ratio $(C_{p_{1/2}}^{8\text{Li}})^2 / (C_{p_{3/2}}^{8\text{Li}})^2$ was taken to be 0.13(2) from Ref. [33]. The optical potential parameters for the entrance and exit channels are listed in Table 1, which were taken from Ref. [30]. Figure 1 displays the DWBA angular distributions of the ${}^7\text{Li}(d, p){}^8\text{Li}_{g.s.}$ reaction with these two sets of optical potential parameters, together with the experimental data. In order to verify if the ${}^7\text{Li}(d, p){}^8\text{Li}$ reaction is peripheral, the ANCs and spectroscopic factors were calculated out using the different geometry parameters of the Woods-Saxon potential for the single-particle bound state with the Set 1 optical potential, as shown in Figure 2. We can find that the spectroscopic factors vary significantly, while the ANCs are nearly a constant, indicating the ${}^7\text{Li}(d, p){}^8\text{Li}_{g.s.}$ reaction is dominated by peripheral process at the present energy.

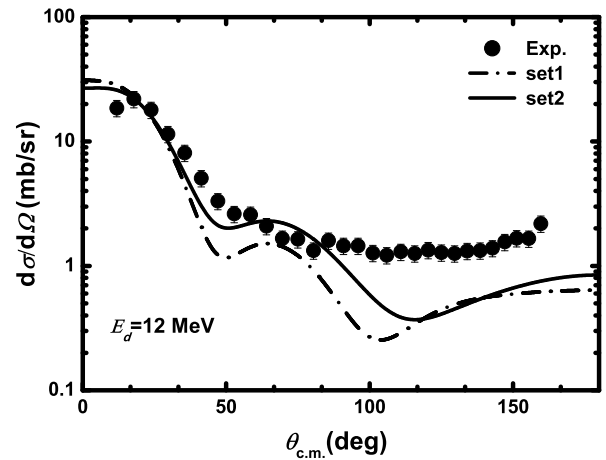


Figure 1 Angular distributions of ${}^7\text{Li}(d, p){}^8\text{Li}_{g.s.}$ at $E_d = 12$ MeV.

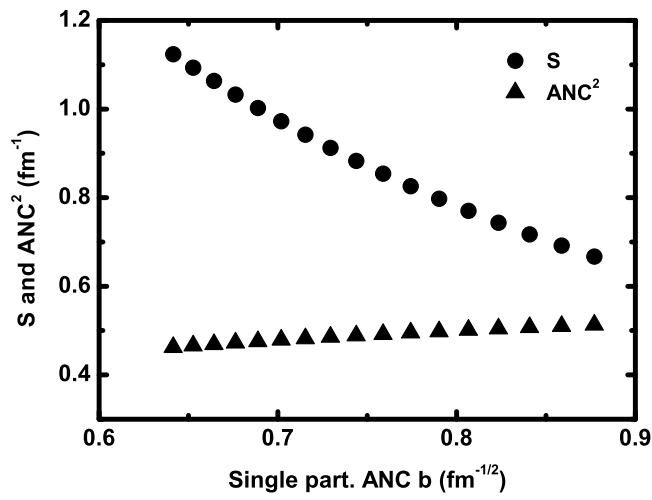


Figure 2 Dependence of the spectroscopic factor (S) and the square of ANC (C^2) for $1p_{3/2}$ orbit of ${}^8\text{Li}$ ground state on the single particle ANC.

The ANC for the virtual decay ${}^8\text{Li}_{g.s.} \rightarrow {}^7\text{Li} + n$ is extracted to be $0.55 \pm 0.097 \text{ fm}^{-1/2}$ using these two sets of optical potential parameters, as listed in Table 2. The error arises from the experiment, the optical potential parameters and the $(C_{p1/2}^{8\text{Li}})^2 / (C_{p3/2}^{8\text{Li}})^2$ ratio.

Table 2 The ANCs for the virtual decay ${}^8\text{Li}_{g.s.} \rightarrow {}^7\text{Li} + n$.

Set NO.	ANC ² (fm ⁻¹)		
	$1p_{3/2}$	$1p_{1/2}$	Total
1	0.49 ± 0.083	0.064 ± 0.011	0.56 ± 0.094
2	0.48 ± 0.089	0.062 ± 0.012	0.54 ± 0.1
average	0.49 ± 0.086	0.063 ± 0.011	0.55 ± 0.097

3 The proton ANC for virtual decay ${}^8\text{B} \rightarrow {}^7\text{Be} + p$

The relationship of the ANCs for mirror systems has been established [34, 35], and utilized to study the ${}^8\text{B}(p, \gamma){}^9\text{C}$, ${}^{11}\text{C}(p, \gamma){}^{12}\text{N}$ and ${}^{26}\text{Si}(p, \gamma){}^{27}\text{P}$ reactions [36–38]. The proton ANC of ${}^8\text{B}$ can be extracted from the neutron ANC of ${}^8\text{Li}$. The relationship between the proton ANC of ${}^8\text{B} \rightarrow {}^7\text{Be} + p$ and the neutron ANC of ${}^8\text{Li} \rightarrow {}^7\text{Li} + n$ can be expressed as

$$(C_{l_f j_f}^{8\text{B}})^2 = R(C_{l_f j_f}^{8\text{Li}})^2, \quad (3)$$

where the ratio R can be computed with [34, 35]

$$R = \left| \frac{F_l(iK_p R_N)}{K_p R_N j_l(ik_n R_N)} \right|^2, \quad (4)$$

where F_l and j_l are the regular Coulomb wave function and the spherical Bessel function, respectively. The wave numbers of proton and neutron k_p (k_n) are determined by the proton (neutron) separation energy ε_p (ε_n) via $k = (2\mu\varepsilon/\hbar^2)^{1/2}$.

On the other hand, if we assume that the single-particle spectroscopic factors S_p and S_n are approximately equal for

mirror system, the ANC ratio R can also be derived with based on the relationship of spectroscopic factor and ANC, $C_{p(n)} = \sqrt{S_{p(n)}} b_{p(n)}$

$$R = \frac{(b_{l_f j_f}^{8\text{B}})^2}{(b_{l_f j_f}^{8\text{Li}})^2}, \quad (5)$$

where the single-particle ANCs $b_{l_f j_f}^{8\text{B}}$ and $b_{l_f j_f}^{8\text{Li}}$ can be derived from the single-particle wave functions calculated with the optical potential model. The ratio R is then extracted to be 1.13 and 1.03 from equations 4 and 5, respectively. We use the same geometry parameters r_0 and a , the same spin-orbit interaction for Woods-Saxon potential, and the same depths adjusted to reproduce the neutron or proton binding energy in ${}^8\text{Li}$ or ${}^8\text{B}$.

The average of above two ratios is derived to be 1.08 ± 0.07 . The squared ANCs for ${}^8\text{B} \rightarrow {}^7\text{Be} + p$ are then derived to be $(C_{p3/2}^{8\text{B}})^2 = 0.53 \pm 0.092 \text{ fm}^{-1}$, $(C_{p1/2}^{8\text{B}})^2 = 0.068 \pm 0.012 \text{ fm}^{-1}$ and $(C_{total}^{8\text{B}})^2 = 0.60 \pm 0.10 \text{ fm}^{-1}$. The error results from the uncertainties of the ANC $(C_{l_f j_f}^{8\text{Li}})^2$ and the ratio R .

4 The astrophysical S-factor and reaction rate of ${}^7\text{Be}(p, \gamma){}^8\text{B}$

At the temperatures of astrophysical interest, the direct capture into the ${}^8\text{B}$ ground state proceeds mainly by the E1 transition from the incoming s- and d-waves to the bound p state. In the present work, the direct capture cross sections and astrophysical S factors were computed based on the measured ANC by using the RADCAP code [39], which is a potential model tool for direct capture reactions. The resulting direct astrophysical S factors as a function of $E_{c.m.}$ are shown in Figure 3. The direct capture S-factor $S_{17}(0)$ at zero energy is then founded to be $20.5 \pm 3.6 \text{ eV b}$. In Figure 4 we compare the present $S_{17}(0)$ factor with the recommended values [1, 47], as well as the previous results from direct measurements [8, 10, 43, 44], transfer reactions [19–21], Coulomb dissociation experiments [15, 18, 45] and charge symmetry method [33, 46]. One can see that the present result agrees with the most recent recommended values, and presents an independent examination to the preexisting results of the ${}^7\text{Be}(p, \gamma){}^8\text{B}$ reaction.

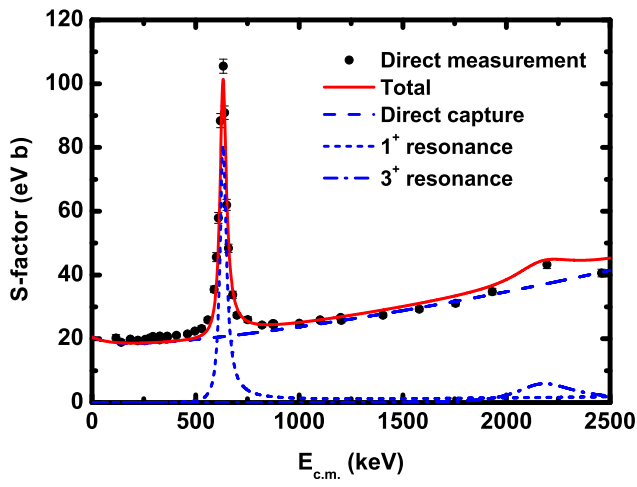


Figure 3 (Color online) Astrophysical S factor for the ${}^7\text{Be}(p, \gamma){}^8\text{B}$ reaction. The solid diamonds denote the experimental data from Ref. [43].

我们的工作，还要引用日本用我们数据的处理！

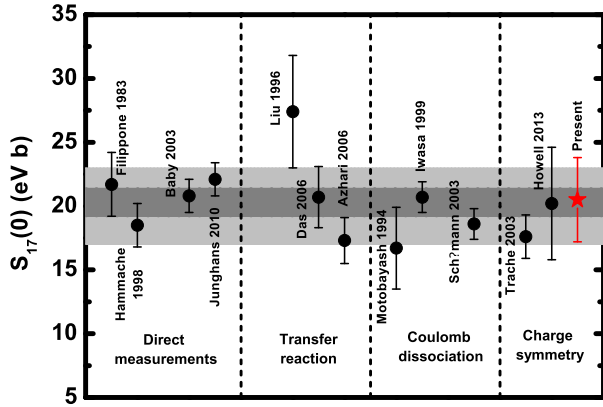


Figure 4 (Color online) Comparison of the present $S_{17}(0)$ factor with the previous results. The direct measurements are performed by Filippone et al. [8], Hammache et al. [44], Baby et al. [10], and Junghans et al. [43]. The transfer reaction was carried out by Liu et al. [19], Das et al. [20] and Azhari et al. [21]. The Coulomb dissociation experiments are performed by Motobayash et al. [45], Iwasa et al. [15], Schumann et al. [18]. The work via charge symmetry method includes the ${}^{13}\text{C}({}^7\text{Li}, {}^8\text{Li}){}^{12}\text{C}$ reaction by Trache et al. [33], the ${}^7\text{Li}({}^8\text{Li}, {}^7\text{Li}){}^8\text{Li}$ reaction by Howell et al. [46], and the ${}^7\text{Li}(\text{d}, \text{p}){}^8\text{Li}$ reaction in the present work (the red star). The light grey and the dark grey areas represent the recommended values by Eric et al. (1998) [47] and by Adelberger et al. (2011) [1], respectively.

The S-factor for the captures via the 1^+ and 3^+ resonances are calculated with Breit-Wigner formula for a single-level resonance,

$$S_{\text{res}}(E) = \pi \frac{(\hbar c)^2}{2\mu c^2} \frac{2J+1}{(2J_1+1)(2J_2+1)} \frac{\Gamma_p(E)\Gamma_\gamma(E)}{(E-E_R)^2 + (\Gamma_{\text{tot}}/2)^2} \exp\left(\frac{E_G}{E}\right)^{1/2}, \quad (6)$$

where E_R represents the resonance energy, J , J_1 and J_2 are the spins of ${}^8\text{B}$, proton and ${}^7\text{Be}$, respectively; Γ_p , Γ_γ and Γ_{tot} are the proton-, gamma- and total widths. The energy depen-

dence of these widths is given by [40]

$$\Gamma_p(E) = \Gamma_p(E_R) \frac{\exp[-(E_G/E)^{1/2}]}{\exp[-(E_G/E_R)^{1/2}]}, \quad (7)$$

and

$$\Gamma_\gamma(E) = \Gamma_\gamma(E_R) \frac{(Q+E)^{2l+1}}{(Q+E_R)^{2l+1}}, \quad (8)$$

where Q is the reaction Q value and l represents the multipolarity of the γ transition. The widths used in our calculation are $\Gamma_p(E_R) = 37 \pm 5$ keV and $\Gamma_\gamma(E_R) = 25 \pm 4$ meV [9] for the 1^+ resonance and $\Gamma_p(E_R) = 350 \pm 40$ keV [41] and $\Gamma_\gamma(E_R) = 150 \pm 30$ meV [42] for the 3^+ resonance. The S-factors of the 1^+ and 3^+ resonances are then derived, as shown in Figure 3. 这里要定量的说一下1+和3+合起来的影响在%几以下！

The astrophysical ${}^7\text{Be}(p, \gamma){}^8\text{B}$ reaction rates for the direct capture and the broad 3^+ resonance can be calculated with

$$N_A < \sigma v >_{nr} = N_A \left(\frac{8}{\pi\mu}\right)^{1/2} \frac{1}{(kT)^{3/2}} * \int_0^\infty S(E) \exp\left[-\left(\frac{E_G}{kT}\right)^{1/2} - \frac{E}{kT}\right] dE, \quad (9)$$

where N_A is Avogadro's number. For the narrow 1^+ resonance, the reaction rate can be approximated by

$$N_A \langle \sigma v \rangle_{\text{res}} = N_A \left(\frac{2\pi}{\mu kT}\right)^{3/2} \hbar^2 \omega \gamma \exp(-E_R/kT), \quad (10)$$

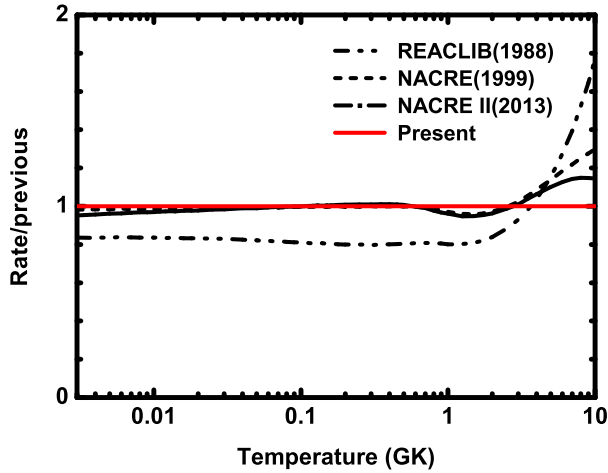
where the resonance strength $\omega\gamma$ is given by

$$\omega\gamma = \frac{2J+1}{(2J_1+1)(2J_2+1)} \frac{\Gamma_p(E_R)\Gamma_\gamma(E_R)}{\Gamma_{\text{tot}}}. \quad (11)$$

The present total reaction rates as a function of temperature are listed in Table 4. In Figure 5 we compare the present rate with the previous compilations of REACLIB [48], NACRE [49] and NACRE II [50]. The present rate is about 20% slower than the REACLIB rate, while is in good agreement with the NACRE rate and the NACRE II rate.

Table 3 The present ${}^7\text{Be}(p,\gamma){}^8\text{B}$ reaction rate in $\text{cm}^3\text{mol}^{-1}\text{s}^{-1}$

T_9	Present	T_9	Present	T_9	Present	T_9	Present
0.003	1.57E-24	0.018	3.72E-11	0.14	2.40E-03	1	1.05E+01
0.004	8.69E-22	0.02	1.34E-10	0.15	3.63E-03	1.25	2.24E+01
0.005	7.71E-20	0.025	1.73E-09	0.16	5.24E-03	1.5	3.82E+01
0.006	2.35E-18	0.03	1.21E-08	0.18	1.00E-02	1.75	5.70E+01
0.007	3.58E-17	0.04	2.04E-07	0.2	1.73E-02	2	7.78E+01
0.008	3.39E-16	0.05	1.50E-06	0.25	5.22E-02	2.5	1.23E+02
0.009	2.26E-15	0.06	6.86E-06	0.3	1.22E-01	3	1.70E+02
0.01	1.15E-14	0.07	2.29E-05	0.35	2.37E-01	3.5	2.19E+02
0.011	4.80E-14	0.08	6.19E-05	0.4	4.10E-01	4	2.69E+02
0.012	1.69E-13	0.09	1.43E-04	0.45	6.50E-01	5	3.77E+02
0.013	5.22E-13	0.1	2.93E-04	0.5	9.66E-01	6	4.95E+02
0.014	1.44E-12	0.11	5.47E-04	0.6	1.87E+00	7	6.22E+02
0.015	3.62E-12	0.12	9.51E-04	0.7	3.21E+00	8	7.59E+02
0.016	8.40E-12	0.13	1.56E-03	0.8	5.09E+00	9	9.04E+02
				0.9	7.64E+00	10	1.05E+03

**Figure 5** (Color online) Comparison of the present reaction rate with the previous compilations of REACLIB [48], NACRE [49] and NACRE II [50].

For convenience, we fit our rate with the expression used in the astrophysical reaction rate library REACLIB [51]

$$\begin{aligned}
 N_A = & \exp[35.5623 - 2.30739T_9^{-1} - 34.3707T_9^{-1/3} \\
 & + 1.57691T_9^{1/3} + 0.909472T_9 - 0.061405T_9^{5/3} \\
 & - 9.58017\ln(T_9)] + \exp[2.42 + 0.0137565T_9^{-1} \\
 & - 12.9036T_9^{-1/3} + 14.9925T_9^{1/3} - 2.55897T_9 \\
 & + 0.0132034T_9^{5/3} - 4.51701\ln(T_9)],
 \end{aligned} \quad (12)$$

The fitting errors are smaller than 1% over the range from $T_9 = 0.01$ to $T_9 = 10$.

5 Summary and conclusion

In summary, we reanalyze the angular distributions of the ${}^7\text{Li}(d,p){}^8\text{Li}$ reaction leading to the ground state of ${}^8\text{Li}$, and derive the neutron ANC of the virtual decay ${}^8\text{Li} \rightarrow {}^7\text{Li} + n$. According to charge symmetry, the proton ANC of ${}^8\text{B} \rightarrow {}^7\text{Be} + p$ is extracted and used to compute the S-factor and the reaction rate of the ${}^7\text{Be}(p,\gamma){}^8\text{B}$ reaction. The new rate is in good agreement with the most recent recommended values. This work provides an independent examination to the existing results for this reaction.

This work was performed with the support of the NSFC under Grant Nos. 11321064, 11490560, 11475264, 11375269 and 11275272, the 973 program of China under Grant No. 2013CB834406.

- Adelberger E G, Garcia A, Hamish R, et al. Solar fusion cross sections. II. The pp chain and CNO cycles. *Rev Mod Phys*, 2011, 83: 195-245
- Haxton W C, Robertson R G H, Serenelli A M, et al. Solar Neutrinos: Status and Prospects. *Ann Rev Astron Astrophys*. 2013, 51: 21-61
- Ahmed S N, Anthony A E, E W Beier, et al. Measurement of the total active ${}^8\text{B}$ solar neutrino flux at the sudbury neutrino observatory with enhanced neutral current sensitivity. *Phys Rev Lett*, 2004, 92: 181301
- Bahcall J N, Pinsonneault M H. What do we (not) know theoretically about solar neutrino fluxes. *Phys Rev Lett*, 2004, 92: 121301
- Bromm V, Larson R B. The first stars. *Annu Rev Astron Astrophys*, 2004, 42: 79-118
- Fuller G M, Woosley S E, Weaver T A. The evolution of radiation-dominated stars. I. nonrotating supermassive stars. *Astrophys J*, 1986, 307: 675-686
- Wiescher M, Görres J, Graff S, et al. The hot proton-proton chains in low-metallicity objects. *Astrophys J*, 1989, 343: 352-364
- Filippone B W, Elwyn A J, Davids C N, et al. Measurement of the ${}^7\text{Be}(p,\gamma){}^8\text{B}$ reaction cross section at low energies. *Phys Rev Lett*, 1983, 50: 412-416
- Filippone B W, Elwyn A J, Davids C N, et al. Proton capture cross section of ${}^7\text{Be}$ and the flux of high energy solar neutrinos. *Phys Rev C*, 1983, 28: 2222-2229
- Baby L T, Bordeanu C, Goldring G, et al. Precision measurement of the ${}^7\text{Be}(p,\gamma){}^8\text{B}$ cross section with an implanted ${}^7\text{Be}$ target. *Phys Rev Lett*, 2003, 90: 022501
- Junghans A R, Mohrmann E C, Snover K A, et al. Precise measurement of the ${}^7\text{Be}(p,\gamma){}^8\text{B}$ S-factor. *Phys Rev C*, 2003, 68: 065803
- Terrasi F, Gialanella L, Imbriani G, et al. Direct measurement of the absolute cross section of $p({}^7\text{Be},\gamma){}^8\text{B}$. *Nucl Phys A*, 2001, 688: 539-542
- Gialanella L, Strieder F, Campajola L, et al. Absolute cross section of $p({}^7\text{Be},\gamma){}^8\text{B}$ using a novel approach. *Eur Phys J A*, 2000, 7: 303-305
- Baur G, Bertulani C A, Rebel H. Coulomb dissociation as a source of information on radiative capture processes of astrophysical interest. *Nucl Phys A*, 1986, 458: 188-204
- Iwasa N, Bou F, Surwka G, et al. Measurement of the Coulomb dissociation of ${}^8\text{B}$ at 254 MeV/nucleon and the ${}^8\text{B}$ solar neutrino flux. *Phys Rev Lett*, 1999, 83: 2910-2913
- Davids B, Anthony D W, Aumann T, et al. $S_{17}(0)$ determined from the Coulomb breakup of 83 MeV/nucleon ${}^8\text{B}$. *Phys Rev Lett*, 2001, 86: 2750-2753
- Schumann F, Hammache F, Typel S, et al. Low-energy cross section of the ${}^7\text{Be}(p,\gamma){}^8\text{B}$ solar fusion reaction from the Coulomb dissociation of ${}^8\text{B}$. *Phys Rev C*, 2006, 73: 015806

- 18 Schumann F, Hammache F, Typel S, et al. Coulomb dissociation of ^8B and the low-energy cross section of the $^7\text{Be}(p,\gamma)^8\text{B}$ solar fusion reaction. *Phys Rev Lett*, 2003, 90: 232501
- 19 Liu W P, Bai X X, Zhou S H, et al. Angular distribution for the $^7\text{Be}(d,n)^8\text{B}$ reaction at $E_{c.m.} = 5.8$ MeV and the $S_{17}(0)$ factor for the $^7\text{Be}(p,\gamma)^8\text{B}$ reaction. *Phys Rev Lett*, 1996, 77: 611-614
- 20 Das J J, Datar M, Sugathan P, et al. Astrophysical $S_{17}(0)$ factor from a measurement of the $^2\text{H}(^7\text{Be},^8\text{B})n$ reaction at $E_{c.m.} = 4.5$ MeV. *Phys Rev C*, 2006, 73: 015808
- 21 Azhari A, Burjan V, Gagliardi C A, et al. Asymptotic normalization coefficients and the $^7\text{Be}(p,\gamma)^8\text{B}$ astrophysical S factor. *Phys Rev C*, 2006, 63: 055803
- 22 Barker F C. The low-energy $^7\text{Be}(p,\gamma)^8\text{B}$ cross section from an R-matrix approach. *Nucl Phys A*, 1995, 588: 693-705
- 23 Robertson G H. Proton Capture by ^7Be and the solar neutrino problem. *Phys Rev C*, 1973, 7: 543-547
- 24 Typel S, Wolter H H, Baur G, et al. Higher-order effects in the Coulomb dissociation of ^8B into $^7\text{Be} + p$. *Nucl Phys A*, 1997, 613: 147-164
- 25 Descouvemont P, Baye D. Microscopic study of the $^7\text{Li}(n,\gamma)^8\text{Li}$ and $^7\text{Be}(p,\gamma)^8\text{B}$ reactions in a multiconfiguration three-cluster model. *Nucl Phys A*, 1994, 567: 341-353
- 26 Csoto A, Langanke K, Koonin S E, et al. $^7\text{Be}(p,\gamma)^8\text{B}$ cross section and the properties of ^7Be . *Phys Rev C*, 1995, 52: 1130-1133
- 27 Descouvemont P. Reanalysis of the $^7\text{Be}(p,\gamma)^8\text{B}$ S-factor in a microscopic model. *Phys Rev C*, 2004, 70: 065802
- 28 Navrátil P, Bertulani C A, Caurier E. $^7\text{Be}(p,\gamma)^8\text{B}$ S-factor from ab initio no-core shell model wave functions. *Phys Rev C*, 2006, 73: 065801
- 29 Petr N, Robert R, Sofia Q, et al. Ab initio many-body calculation of the $^7\text{Be}(p,\gamma)^8\text{B}$ radiative capture. *Phys Lett B*, 2011, 704: 379-383
- 30 Schiffer J P, Morrison G C, Siemssen R H. Study of the (d,p) reaction in the $1p$ Shell. *Phys Rev*, 1967, 83: 195-245
- 31 Thompson I J. Coupled reaction channels calculations in nuclear physics. *Comput Phys Rep*, 1988, 7: 167
- 32 Blokhintsev L D, Borbely I, Dolinskii E I. Nuclear vertex constants. *Sov J Part Nucl*. 1977, 8: 485
- 33 Trache L, Azhari A, Carstoiu F, et al. Asymptotic normalization coefficients for $^8\text{B} \rightarrow ^7\text{Be} + p$ from a study of $^8\text{Li} \rightarrow ^7\text{Li} + n$. *Phys Rev C*, 2003, 67: 062801
- 34 Timofeyuk N K, Johnson R C, Mukhamedzhanov A M. Relation between proton and neutron asymptotic normalization coefficients for light mirror nuclei and its relevance to nuclear astrophysics. *Phys Rev Lett*, 2003, 91: 232501
- 35 Timofeyuk N K, Descouvemont P. Asymptotic normalization coefficients for mirror virtual nucleon decays in a microscopic cluster model. *Phys Rev C*, 2005, 71: 064305
- 36 Guo B, Li Z H, Liu W P, et al. The $^8\text{Li}(d,p)^9\text{Li}$ reaction and astrophysical $^8\text{B}(p,\gamma)^9\text{C}$ reaction rate. *Nucl Phys A*, 761: 162-172
- 37 Guo B, Li Z H, Bai X X, et al. Determination of the astrophysical $^{26}\text{Si}(p,\gamma)^{27}\text{P}$ reaction rate from the asymptotic normalization coefficients of $^{27}\text{Mg} \rightarrow ^{26}\text{Mg} + n$. *Phys Rev C*, 2006, 73: 048801
- 38 Guo B, Li Z H, Liu W P, Bai X X. Determination of astrophysical $^{11}\text{C}(p,\gamma)^{12}\text{N}$ reaction rate from the asymptotic normalization coefficients of $^{12}\text{B} \rightarrow ^{11}\text{B} + n$. *J Phys G*, 2007, 34: 103-114
- 39 Bertulani C A. RADCAP: A potential model tool for direct capture reactions. *Comput Phys Commun*, 2003, 156: 123-141
- 40 Rolfs C E, Rodney W S. *Cauldrons in the cosmos*, University of Chicago Press, 1988
- 41 Ajzenberg-Selove F. Energy level of light nuclei $A = 5 - 10$. *Nucl Phys A*, 1988, 490: 1-225
- 42 Baby L T, Bordeanu C, Goldring G, et al. New measurement of the proton capture rate on ^7Be and the $S_{17}(0)$ factor. *Phys Rev C*, 2003, 67: 065805
- 43 Junghans A R, Snover K A, Mohrmann E C, et al. Updated S factors for the $^7\text{Be}(p,\gamma)^8\text{B}$ reaction. *Phys Rev C*, 2010, 81: 012801
- 44 Hammache F, Bogaert G, Aguer P, et al. New measurement and analysis of the $^7\text{Be}(p,\gamma)^8\text{B}$ cross section. *Phys Rev Lett*, 1998, 80: 928-931
- 45 Motobayashi T, Iwasa N, Ando Y, et al. Coulomb dissociation of ^8B and the $^7\text{Be}(p,\gamma)^8\text{B}$ reaction at low energies. *Phys Rev Lett*, 1994, 73: 2680-2683
- 46 Howell D, Davids B, Greene J P, et al. First determination of the ^8Li valence neutron asymptotic normalization coefficient using the $^7\text{Li}(^8\text{Li},^7\text{Li})^8\text{Li}$ reaction. *Phys Rev C*, 2013, 88: 025804
- 47 Eric G A, Sam M A, John N B, et al. Solar fusion cross sections. *Rev Mod Phys*, 1998, 70: 1265-1291
- 48 Georgeanne R C, Willam A F. Thermonuclear reaction rates V. Atomic data and nuclear data tables, 1988, 40: 283-334
- 49 Angulo C, Arnould M, Rayet M, et al. A compilation of charged-particle induced thermonuclear reaction rates. *Nucl Phys A*, 1999, 656: 3-183
- 50 Xu Y, Takahashi K, Goriely S, et al. An update of the NACRE compilation of charged-particle-induced thermonuclear reaction rates for nuclei with mass number $A < 16$. *Nucl Phys A*, 2013, 918: 61-169
- 51 Rauscher T, Thielemann F K. Tables of Nuclear Cross Sections and Reaction Rates: An addendum to the paper "astrophysical reaction rates from statistical model calculations". *At. Data Nucl. Data Tables*, 2001, 79: 47-64